

FFT-Accelerated Worldline Monte Carlo for Fermionic Lattice Models

A Determinant-Free Approach with $O(N \log N)$ Complexity

Deqian Kong¹, Shi Feng^{2,3}, Jianwen Xie⁴, Ying Nian Wu¹

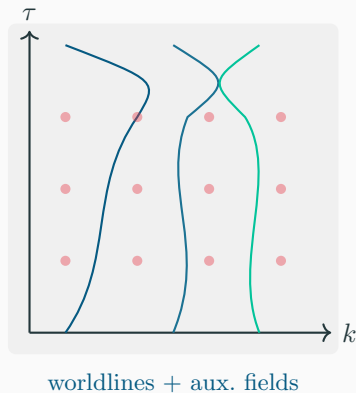
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Outline

1. The computational bottleneck in quantum simulation
2. Key idea: joint sampling of worldlines & auxiliary fields
3. Momentum-space transfer kernel & convolution structure
4. **Two core MCMC moves:**
 - FFBS: Forward Filtering Backward Sampling for worldlines
 - Permutations: resample worldline topology to recover fermion statistics
5. Fast auxiliary field updates
6. Benchmark on 2D Hubbard model



Current status

- Half filling on bipartite lattices: unbiased finite- T benchmarks are now high-precision (PRL **134**, 016503 (2025); PRB **111**, 035123 (2025)).
- Besides the sign problem, core bottleneck is still determinant linear algebra; fast-update/submatrix schemes reduce prefactors.

Method	Cost per sweep	Representative refs
BSS-DQMC	$O(\beta N^4)$	Blankenbecler <i>et al.</i> , PRD 24 , 2278 (1981); D. Scalapino, R. Sugar, PRB 24 , 4295 (1981)
local HS flips	$O(\beta N^3)$	Hirsch, PRB 31 , 4403 (1985)
Delayed/submatrix batching	Similar asymptotic scaling, smaller prefactor	Nukala <i>et al.</i> , PRB 80 , 195111 (2009); Sun, Xu, PRB 109 , 235140 (2024)
FFT-FFBS	$O(\beta N \log N)$	D. Kong, et al., arXiv:2510.13866

The Determinant Bottleneck in DQMC

Standard approach (Blankenbecler, Scalapino, Sugar 1981):

1. Trotter decompose: $e^{-\beta H} = \prod_{\ell} e^{-\Delta\tau H}$
2. HS-decouple: introduce auxiliary fields $\Sigma = \{s_{\mathbf{r}\ell}\}$ to linearize
3. **Integrate out fermions analytically** $\Rightarrow$ determinant

$$Z = \sum_{\Sigma} \underbrace{\prod_{\sigma} \det M_{\sigma}(\Sigma)}_{O(N^2) \sim O(N^3) \text{ per update}} e^{-S_{\text{aux}}(\Sigma)}$$

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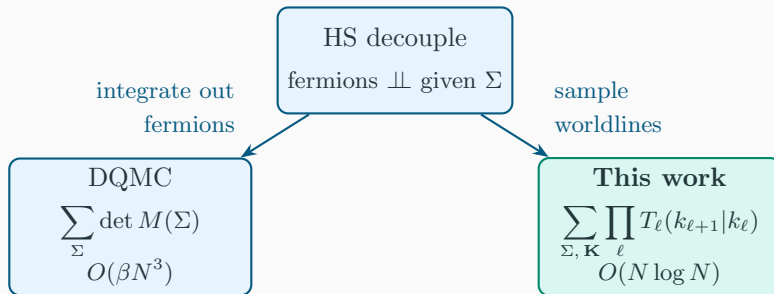
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Key Idea: Don't integrate out the fermions. Sample k -space worldlines *jointly* with HS field (Σ).

Key Idea: Joint Worldline–Auxiliary Field Sampling



Once decoupled by Σ , each fermion is **independent**:

$$\pi(\mathbf{K}_{\uparrow}, \mathbf{K}_{\downarrow}, \Sigma, P_{\uparrow}, P_{\downarrow}) \propto P_{\text{HS}}(\Sigma) \cdot \text{sgn}(P_{\uparrow}) \text{sgn}(P_{\downarrow}) \cdot \prod_{\sigma, p, \ell} \left\langle k_{\ell+1, \sigma}^{(p)} \left| T_{\ell, \sigma}[\Sigma] \right| k_{\ell, \sigma}^{(p)} \right\rangle$$

Three-component Gibbs:

$$\mathbf{K} \mid \Sigma \text{ (FFBS)} \longleftrightarrow \Sigma \mid \mathbf{K} \text{ (exact conditionals)} \longleftrightarrow P_{\sigma} \text{ (permutation moves)}$$

The Transfer Kernel and Convolution by FFT in k -Space

Fermi Hubbard model:

$$H = K + U \sum_{\mathbf{r}} \left(n_{\mathbf{r}\uparrow} - \frac{1}{2} \right) \left(n_{\mathbf{r}\downarrow} - \frac{1}{2} \right), \quad K = -t \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} (c_{\mathbf{r}}^{\dagger} c_{\mathbf{r}'} + \text{h.c.})$$

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$$T_{\ell} = e^{-\frac{\Delta\tau}{2}K} e^{-\Delta\tau V_{\ell}} e^{-\frac{\Delta\tau}{2}K}$$

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Insert $\sum_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}|$ resolutions of identity twice:

$$\langle k' | T_\ell | k \rangle = \frac{1}{V} \underbrace{e^{-\frac{\Delta\tau}{2}\varepsilon_{k'}}}_{D(k')} \underbrace{\sum_{\mathbf{r}} W_\ell(\mathbf{r}) e^{i(k-k')\cdot\mathbf{r}}}_{\widehat{W}_\ell(k-k') \leftarrow \text{DFT of } W_\ell} \underbrace{e^{-\frac{\Delta\tau}{2}\varepsilon_k}}_{D(k)}$$

$$\varepsilon_k = -2t \sum_{\alpha} \cos k_{\alpha}, \quad W_\ell(\mathbf{r}) = e^{-\Delta\tau V_\ell(\mathbf{r})}.$$

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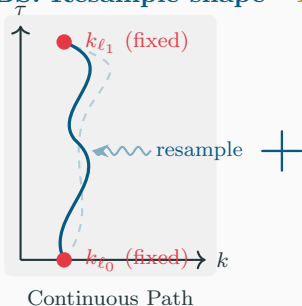
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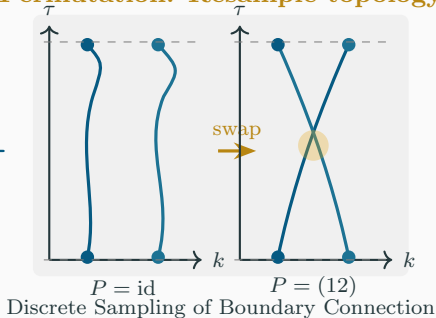
Dependence on $k - k' \implies$ **convolution** $\implies$ **FFT** in $O(N \log N)$

Two Core MCMC Moves for Worldlines: Path and Permutation

FFBS: Resample shape



Permutation: Resample topology



Both are essential for ergodicity. FFBS explores continuous momenta along each worldline. Permutation moves explore how worldlines connect at $\tau = \beta$, encoding $\text{sgn}(P)$.

Block Update for Worldlines by Forward Filtering Backward Sampling

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- Fix k_{ℓ_0}, k_{ℓ_1} and update the whole interior block.

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stores weights of all partial paths (exact). Sample interior paths with probability proportional to the weight.

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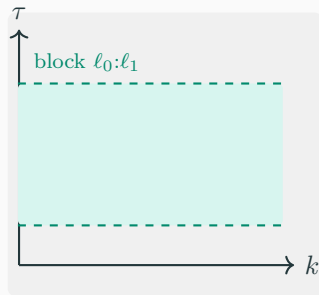
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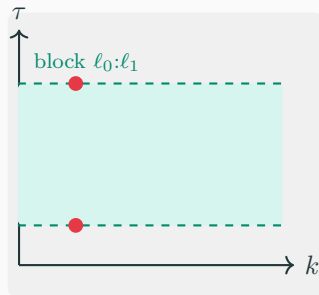
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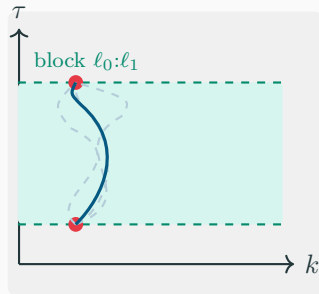
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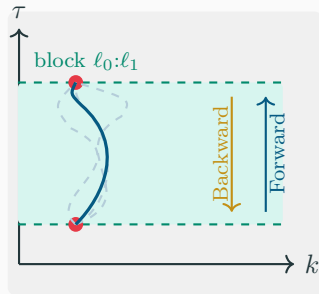
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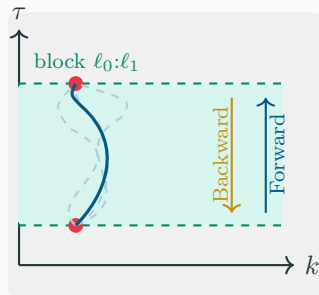
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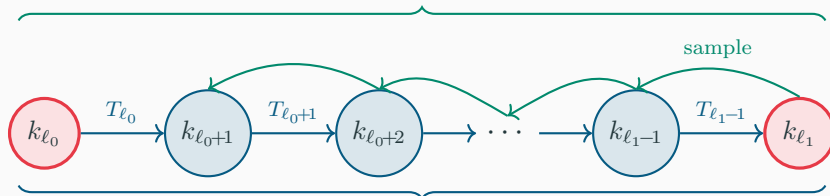


Forward: total weight of all partial paths to slice ℓ ending at k .
Backward: sample full interior paths from those weights.

Forward Filtering Backward Sampling (FFBS): Block Sampling of Worldlines

Goal: Given fixed endpoints k_{ℓ_0} , k_{ℓ_1} , resample $\{k_{\ell_0+1}, \dots, k_{\ell_1-1}\}$ from the *exact* conditional.

Backward: draw $k_\ell \sim P(k_\ell \mid k_{\ell+1})$



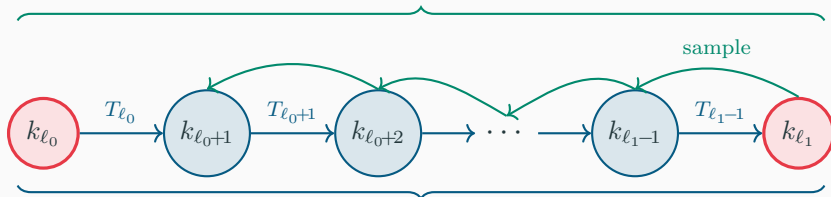
Forward: compute $\alpha_\ell(k)$ via FFT convolution

1. **Forward:** Propagate $\alpha_{\ell+1}(k) = D(k) \sum_{k'} \frac{\widehat{W}_\ell(k-k')}{V} D(k') \alpha_\ell(k')$
[convolution $\Rightarrow$ FFT $\sim O(N \log N)$]

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[convolution $\Rightarrow$ FFT $\sim O(N \log N)$]
2. **Backward:** Draw k_ℓ from $P(k_\ell \mid k_{\ell+1}) \propto \alpha_\ell(k_\ell) \cdot \langle k_{\ell+1} | T_\ell | k_\ell \rangle$

This is an **exact Gibbs step**: draws from the true conditional $P(\text{block} \mid \text{endpoints}, \Sigma)$.

FFBS Block Placement: Scramble End Points

FFBS conditions on block endpoints $\Rightarrow$ endpoints not resampled within that block.

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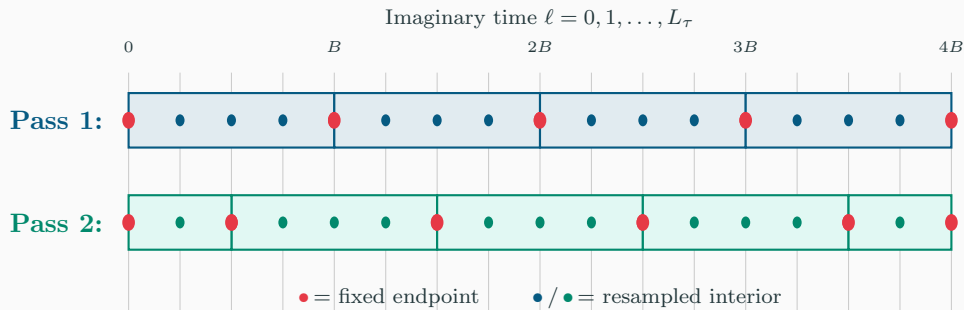
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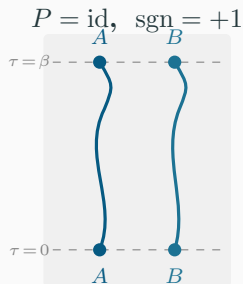
Requirement: Every time slice must appear as an interior point in at least one block. **Simplest solution:** Two passes offset by $B/2$.



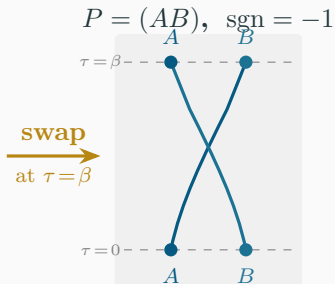
Any scheme satisfying the coverage requirement works (random blocks, sliding windows, etc.).

Permutation Updates for Fermionic Statistics

FFBS resamples momenta along *each* worldline, but never changes *which worldline connects to which* at $\tau = \beta$. The fermionic partition function requires summing over permutations weighted by $\text{sgn}(P)$.



Each closes on itself



Worldlines exchange endpoints

swap
at $\tau = \beta$

Note:

FFBS *never* reconnects worldline endpoints.

Without permutation moves:

$\Rightarrow$ stuck in identity sector

$\Rightarrow$ miss all fermionic physics

Implementation:

Propose pair swap at $\tau = \beta$.

Accept/reject by Metropolis.

Auxiliary Field Updates: Cut-Invariant Form

Note: sampled k -space worldlines provide only $n_k(\ell)$ (momentum-basis diagonal), not directly $n(\mathbf{r}, \ell)$, which depends on off-diagonal elements in k -space. So we use the the following form to get site-wise HS conditionals.

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Real space conditional

$$P(s_{\mathbf{r}\ell} = s \mid \text{rest}) = \frac{w_s}{w_{+1} + w_{-1}}$$

$$w_s = \prod_{\sigma} \left[\Gamma_{\sigma,\ell} + C_{\sigma,\ell}(\mathbf{r}) e^{\sigma \lambda_s} \right]$$

$$\Gamma_{\sigma,\ell} = \sum_{\mathbf{x} \neq \mathbf{r}} C_{\sigma,\ell}(\mathbf{x}) e^{\sigma \lambda_{\mathbf{x}\ell}}$$

Algorithm Summary

1. **Precompute** potential weights $\widehat{W}_{\sigma,\ell}(q) = \sum_{\mathbf{r}} W_{\sigma,\ell}(\mathbf{r}) e^{iq \cdot \mathbf{r}}$ $O(\beta \cdot N \log N)$

Total per-sweep complexity: $O(\beta N \log N)$

Each step satisfies detailed balance $\Rightarrow$ composition preserves $\pi(\mathbf{K}, \Sigma, P)$.

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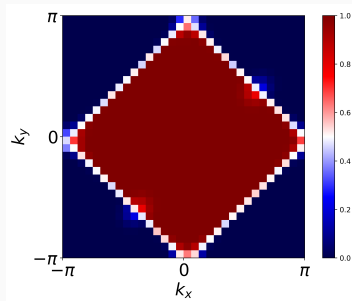
4. **Auxiliary field Gibbs:** sweep $s_{\mathbf{r}\ell}$ via cut-invariant conditionals $O(\beta N \log N)$
Exact Bernoulli sampling at each spacetime point; recompute $C_{\sigma,\ell}$ between slices.

Total per-sweep complexity: $O(\beta N \log N)$

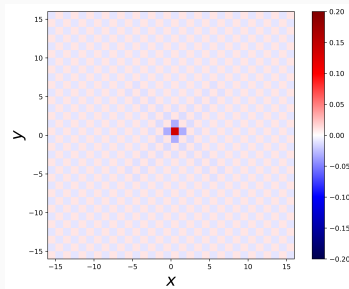
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Results: Weak Coupling ($U/t = 1$, $\beta t = 32$, 32×32)

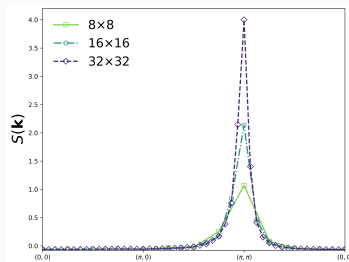
Charge: Sharp Fermi surface in $\langle n_{\mathbf{k}} \rangle$, discontinuity at $\varepsilon(\mathbf{k}) = 0$. **Spin:** Staggered AF $\langle S_0^z S_{\mathbf{r}}^z \rangle$; $S(\mathbf{k})$ peaks at $\mathbf{Q} = (\pi, \pi)$, grows with L .



Momentum distribution $n_{\mathbf{k}}$



Real-space $C(\mathbf{r}) = \langle S_0^z S_{\mathbf{r}}^z \rangle$

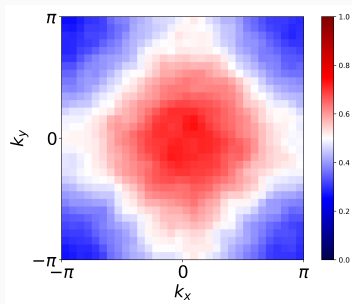


Spin structure factor $S(\mathbf{k})$

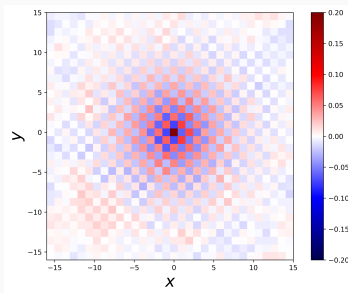
All results match DQMC benchmarks (Hirsch 1985, Tomas et al. 2012).

Results: Strong Coupling ($U/t = 20$, $\beta t = 32$, 32×32)

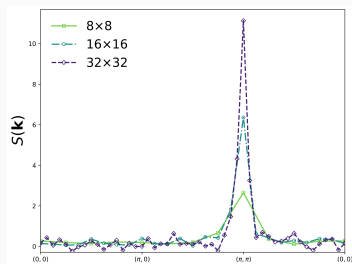
Mott insulator: No sharp FS, $n_{\mathbf{k}} \rightarrow 1$ as $U/t \rightarrow \infty$, double occupancy suppressed. **Heisenberg magnet:** $J_{\text{eff}} = 4t^2/U$; staggered $\langle S_0^z S_{\mathbf{r}}^z \rangle$ with exponential decay; $S(\mathbf{k})$ AF peak at (π, π) grows with L .



Momentum distribution $n_{\mathbf{k}}$



Real-space $C(\mathbf{r}) = \langle S_0^z S_{\mathbf{r}}^z \rangle$



Spin structure factor $S(\mathbf{k})$

Both limits recovered: Fermi liquid $\xrightarrow{U/t}$ Mott insulator. Matches White et al. 1989, Moreo et al. 1990.

Method Comparison

	Fast-Update DQMC	FFT-FFBS
Fermion treatment	Integrate out (det)	Sample worldlines
Per-sweep cost	$O(\beta N^3)$	$O(\beta N \log N)$
Block updates	Expensive	FFBS via FFT
Auxiliary HS field	Metropolis	Exact Gibbs
Parallelization	Limited	Easy GPU Implementation
Long-range interaction	Increases to $O(\beta N^4)$	Remains $O(\beta N \log N)$

The key advantage: replacing N^3 per sweep with $N \log N$ convolutions.

D. Kong, *et al.*, arXiv:2510.13866

Determinant-free QMC with $O(N \log N)$ scaling

Key ingredients:

1. **Joint sampling:** Keep k -space fermion worldlines as explicit degrees of freedom alongside auxiliary fields — avoid the $O(N^2) \sim O(N^3)$ determinant update entirely
2. **FFT-accelerated FFBS:** The k -space transfer kernel is a convolution, enabling exact block updates of worldlines in $O(N \log N)$. Two offset sweeps ensure full ergodicity along each worldline.

Thank you!

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